

Move Argo CD Server to node02

Kubernetes homelab - operational procedure | Target node: node02 (192.168.178.52)

Goal: pin the **argocd-server** Deployment to node02 using a nodeSelector. This does not relocate other Argo CD components or change the NodePort Service.

1. Check cluster health

```
kubectl get nodes -o wide
kubectl get pods -n argocd -o wide
```

Confirm node02 shows **Ready**. It was previously reported NotReady; if that is still true, fix node health before proceeding. Check for scheduling restrictions:

```
kubectl describe node node02 | grep -A8 -E "Taints:|Conditions:"
kubectl get node node02 --show-labels
```

Confirm the hostname label includes **kubernetes.io/hostname=node02** and there are no untolerated taints. Check that node02 has sufficient available resources.

2. Save current Deployment configuration

```
kubectl get deployment argocd-server -n argocd -o yaml \
> argocd-server-before-node02.yaml
```

3. Pin the server to node02

```
kubectl patch deployment argocd-server -n argocd \
--type=merge \
-p '{"spec":{"template":{"spec":{"nodeSelector":
{"kubernetes.io/hostname":"node02"}}}}}'
```

The patch updates the Pod template and triggers a rolling update automatically. No additional rollout restart is needed. If an existing nodeSelector is present, review it first: a merge patch adds/replaces only the hostname key.

4. Wait and verify

```
kubectl rollout status deployment/argocd-server \
-n argocd --timeout=180s
kubectl get pods -n argocd -o wide
kubectl get deployment argocd-server -n argocd \
-o jsonpath="{.spec.template.spec.nodeSelector}"; echo
```

Expected: the new argocd-server Pod shows **Running**, **1/1 Ready**, and **NODE=node02**. Other Argo CD Pods may still run elsewhere.

5. Check permanent browser access

```
kubectl get svc argocd-server -n argocd
```

If the HTTPS NodePort is configured as **30081**, try **https://192.168.178.52:30081**. The Service remains reachable through other healthy node IPs too, depending on firewall and NodePort configuration. The port is not tied to the Pod's node. Confirm the actual allocated NodePort rather than assuming 30081.

6. Troubleshoot a Pending Pod

```
kubectl get pods -n argocd -o wide
```

```
kubectl describe pod -n argocd <NEW_ARGOCD_SERVER_POD>
kubectl describe node node02
```

Look at the Pod Events section for NotReady, insufficient CPU/memory, untolerated taints, or node affinity conflicts. Do not remove taints blindly.

7. Roll back to automatic scheduling

```
kubectl patch deployment argocd-server -n argocd \
  --type=json \
  -p ' [{"op": "remove", "path":
    "/spec/template/spec/nodeSelector/kubernetes.io~1hostname"} ] '
kubectl rollout status deployment/argocd-server -n argocd
```

This removes only the hostname restriction added above, preserving other nodeSelector keys. If the Deployment was originally managed by Helm or GitOps, also update the source configuration; otherwise a future reconciliation may undo this manual change.

Important behavior

Pinning is not high availability: If node02 becomes unavailable, the server cannot automatically move to another node while this selector is present. The Argo CD UI/API may become unavailable, although other Argo CD components may continue operating. For resilient operation, avoid a hard node pin and consider replicas and appropriate scheduling rules.